Ayman **Anwar**

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Summary

PhD Candidate at Electrical & Computer Engineering, University of Toronto, with 7+ years of experience in machine/deep learning and NLP. Extensive background in health informatics, signal processing, and medical imaging. Strong expertise in transformer architectures and LLMs, including training and optimization. Proficient in Python, PyTorch, and C/C++.

Skills

Programming	Python, C/C++, and MATLAB.
AI/ML Frameworks	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, OpenCV, Torchvision.
Computer Science	Algorithms, Data Structures, Computer Vision, Artificial Intelligence, Natural Language Processing.
Other Skills	Frontend(HTML, Bootstrap, CSS3), Backend(Django, Flask), Database (MySQL)

Education

Electrical and Computer Engineering, University of Toronto (UofT) *Toronto, Canada*
PhD candidate *May 2022 - Now*

- **Tentative graduation date**: Nov. 2025.
- **Position**: Graduate RA at AI in Health Outcomes Lab at North York General Hospital (NYGH) Supervised by Prof. Ervin Sejdic.
- **Research**: AI in healthcare applications (Biosignal analyses, Medical imaging, and NLP for clinical notes).

Faculty of Engineering, Cairo University *Cairo, Egypt*
MSc. and BSc. in Systems and Biomedical Engineering *sep. 2012 - Aug. 2020*

- **MSc. Thesis Title**: EEG signal classification using multidimensional CNNs and autoencoders for real-time BCI. **MSc GPA**: 3.9.
- **BSc. Thesis Title**: Automatic Prosthesis Control Using Human Thoughts. **BSc GPA**: 3.8.
- **Research**: Classical ML, DL, Computational Neuroscience, Brain Computer Interface, and Information Theory.

Experience

Innovative Medical Engineering Developments (iMED) Lab. *Toronto, Canada*
Research Assistant | AI | ML | Python | Pytorch | LLMs *May. 2022 - Now*

- Developed and evaluated transformer architectures for classification and segmentation of accelerometer signals.
- Trained and maintained cross-domain GANs, LSGANs, and CycleGANs to enhance model generalization across diverse datasets.
- Designed and implemented imbalance-handling pipelines combining resampling, augmentation, and two-stage training, leading to a significant boost in model performance (e.g., 20% F1-score) on underrepresented classes.
- Supervised multiple projects leveraging NLP and MLLMs for predictive modeling in healthcare applications.

Electrical and Computer Eng. University of Toronto. *Toronto, Canada*
Course Instructor and Teaching Assistant *May. 2022 - Now*

- Course Instructor for ECE244: Programming Fundamentals (Fall 2024); developed lecture notes, labs, and assessments covering core programming concepts in C++.
- Teaching Assistant for ECE1513: Introduction to Machine Learning (Spring 2023 & 2024), APS105: Computer Fundamentals (Spring 2023 & 2024), ECE244: Programming Fundamentals (Fall 2023), and ECE302: Probability and Applications (Fall).

WeCloudData *Toronto, Canada*
AI and machine learning Instructor *September 2023 - December 2024*

- Taught deep learning and computer vision using PyTorch, including CNNs and Vision Transformer fine-tuning.
- Delivered NLP and small LLM training, covering traditional NLP methods and model fine-tuning.
- Designed hands-on Python exercises and notebooks across AI frameworks (PyTorch, TensorFlow) to reinforce practical skills.

Pulse for Integrated Solutions. *Cairo, Egypt*
Machine Learning and DSP Engineer | Python | Tensorflow | C++ *July. 2018 April. 2022*

- Prepared and preprocessed ECG datasets for segmentation and classification of cardiac abnormalities.
- Trained CNN and RNN models for ECG signal segmentation and classification, including AF, VF, and ST-segment deviation.
- Developed ECG delineation and QRS algorithms meeting IEC60601-2-25:2011 and IEC60601-2-47:2012 for CE certification.
- Deployed models in a web application using TensorFlow with a C++ backend for real-time clinical use.
- Fine-tuned and trained BERT models for document classification and entity recognition on clinical notes.
- Developed NLP pipelines to extract patient state, illness, history, and case severity for recommendation systems.

Projects

Clinical Report Generation from 3D CT Scans using Multimodal LLMs *Toronto, Canada*
PYTHON | PYTORCH | LLMS | HUGGING FACE *May 2025 - Now*
- Trained/Finetuned a Vision Transformer on 3D CT volumes using multitask learning for 18 tasks (CT-Rate dataset).
- Applied late fusion to combine ViT embeddings with LLaMA 3.2 prompts for clinical report generation.
- Evaluated performance using ROUGE and BERTScore; currently exploring alternative fusion techniques and multimodal training.

Novel Two-Stage Training with Auxiliary Tasks for Imbalanced Data *Toronto, Canada*
PYTHON | PYTORCH | SKLEARN | SELF-SUPERVISED LEARNING *Jan. 2025 - Aug. 2025*
- Developed a two-stage approach using a transformers: pretraining on self-supervised auxiliary tasks for minority classes.
- Tested in a signal-processing environment to classify real-world biomedical signals for swallowing severity detection.
- Compared performance against resampling, augmentation, and generative-based techniques. underrepresented classes.

Time-series Transformers for Spatial Information Tracking with Multi-task Learning *Toronto, Canada*
PYTHON | PYTORCH | TST TRANSFORMERS *May. 2024 - Oct. 2024*
- Developed a novel time-series transformer architecture to track anatomical structure coordinates as a regression task.
- Implemented hard-parameter sharing and soft-parameter sharing multi-task learning techniques to improve model performance.
- Predicted pixel-level locations from signals with <0.05 error, demonstrating high spatial accuracy for signal data only.
- **Publication**: *Novel Videographic-Free Tracking Anatomical Structures Using Accelerometer Signals and Multi-task Transformers.*

Cross-Domain Medical Signals and Image Translation Using CycleGANs *Toronto, Canada*
PYTHON | TENSORFLOW | TRANSFORMERS | TENSORBOARD *January. 2023 - Oct. 2023*
- Developed a CycleGAN variant combining CNNs and transformer blocks for cross-domain medical image translation.
- Implemented stabilization techniques (e.g., spectral normalization, two-time-scale update rules) to mitigate GAN instability.
- Robust transferability: Performance only dropped by $<5\%$ in accuracy and other metrics when transferred to a different environment.
- **Publication**:*Towards a comprehensive bedside swallow screening protocol using cross-domain transformation and HRCA*.

Automated Skin Burn Segmentation and Severity Classification Using CNNs *Toronto, Canada*
PYTHON | PYTORCH | CNNS | SKLEARN | MATPLOTLIB *Sept. 2023 - Apr. 2024*
- Supervised a student-led project developing a tool to segment skin burns and classify severity from medical images.
- Trained and evaluated multiple CNN architectures (ResNet, EfficientNet, etc.) for improved segmentation and classification.
- Outperformed prior literature by 20% in accuracy and 15% in F1 score.
- **Publication**: *Integrating multi-source data for skin burn classification using deep learning*.

Smartphone-Based Intoxication Detection Using Ensemble ML Models *Toronto, Canada*
PYTHON | SKLEARN | SVMS| CLASSICAL ML MATPLOTLIB *Oct. 2022 - Apr.. 2023*
- Developed a majority-voting ensemble using SVMs and Random Forests to detect intoxication levels from microphone data.
- Achieved $>90\%$ accuracy, outperforming baseline models in real-world testing.
- Collaborated with Stanford University; **Publication**: *Detection of alcohol intoxication using voice features*.

Recent Publications

Non-Invasive Integrated Swallowing Kinematic Analysis Framework Leveraging Transformer-Based Multi-Task Neural Networks *August 2025*
A. ANWAR, ET AL. | ELSEVIER - COMPUTERS IN BIOLOGY AND MEDICINE

Transformers in biosignal analysis: A review *October 2024*
A. ANWAR, ET AL. |ELSEVIER -INFORMATION FUSION

Swallowing assessment using high-resolution cervical auscultations and transformer-based neural networks *July 2024*
A. ANWAR, ET AL. | IEEE EMBC 2024

References

Prof. Ervin Sejdic
PROFESSOR AT ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT, UNIVERSITY OF TORONTO.
- ervin.sejdic@utoronto.ca

PhD. Yassin Khalifa
DATA SCIENTIST AT CENTER FOR RESEARCH COMPUTING AND DATA, UNIVERSITY OF PITTSBURGH.
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